

TSHEDZA TSHIKOVHI

Software Developer | Data Scientist | UI/UX Designer
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PROFESSIONAL SUMMARY

Full-Stack Software Developer and Data Scientist with hands-on experience across the entire product lifecycle: gathering user requirements, designing interfaces and design systems, building frontend and backend systems, and applying machine learning to real-world problems. Currently a Software Development Intern at Shudu Connections (Innovation Hub), where I lead a small team building an SME incubation management platform. Independently designed, built, and deployed multiple live full-stack products for South African users, spanning fintech, mental wellness, and sports analytics. Top-20 finalist in a national business pitch competition. Passionate about applying technology to real-world South African challenges.

ACHIEVEMENTS

- Top-20 Finalist - 2026 Pitch Your Business Competition (North-West University), selected from the entrant pool to advance to Business Model Canvas training
- Featured on SABC 1's Daily Thetha and SABC 2's Our Space as an emerging changemaker, sharing insights on technology, innovation, and digital futures

EXPERIENCE

Software Developer Intern

Shudu Connections - Innovation Hub | 2026 - Present

- Leading a small internship team building TractionLog, an SME incubation management platform (Angular, Firebase, role-based dashboards)
- Developing full-stack and data-driven applications in an innovation-focused environment
- Working on software systems using modern development frameworks and tools
- Collaborating using Git-based workflows in team environments
- Contributing to backend logic, APIs, and UI development
- Supporting product design and system architecture decisions

Senior Fruit & Veg Assistant

Shoprite | Nov 2024 - Present

- Managed inventory control and quality assurance while supporting customers and store operations
- Developed discipline, teamwork, and operational efficiency skills

Student Representative Council (SRC)

Eduvos Midrand Campus | 2023 - 2024

- Represented student body in academic and institutional matters
- Organised student initiatives and campus engagement activities
- Strengthened leadership, communication, and problem-solving skills

TECHNICAL SKILLS

- Languages: Python, SQL, R, Java, JavaScript, TypeScript, HTML, CSS

- Frameworks & Libraries: React.js, Angular 18, Tailwind CSS, Angular Material, Firebase, Streamlit, Pandas, NumPy, Matplotlib, Scikit-learn, ApexCharts
- Backend & APIs: REST APIs, OOP, Firebase (Auth, Firestore, Storage, Cloud Functions), Google Gemini API/Firebase Genkit
- UI/UX Design: Wireframing, Design Systems & Style Guides, Responsive Design, Dashboard & Data Visualisation
- Data Science & ML: Predictive Modelling, Feature Engineering, Statistical Analysis, Machine Learning
- Tools: Git & GitHub, VS Code, Jupyter Notebook, Linux
- Other: Cloud Architecture, Cybersecurity Fundamentals, Generative AI (in progress)

TECHNICAL PROJECTS

TractionLog - SME Incubation Management Platform

Team Lead - Full-Stack Developer & UI/UX Designer | Shudu Connections (Innovation Hub)

- Led a small internship team to design and build a management platform used by incubation hubs, accelerators, and development programmes to track and measure SMME growth
- Directed information architecture across three distinct user portals - centre administrator, SME business owner, and TIA administrator - each with dedicated navigation and Firestore-scoped data access
- Designed and built real-time dashboards (ApexCharts) translating raw programme data into insights on business progress and ecosystem performance
- Implemented role-based access control via Firebase custom JWT claims; built the Firestore service layer, Cloud Functions, and Google Gemini integration (via Firebase Genkit)
- Tools: Angular 18, Firebase 10 (Auth, Firestore, Storage, Cloud Functions), Angular Material, ApexCharts, Google Gemini/Firebase Genkit

CashCram - Student Financial Wellness Platform (B2B2C SaaS)

Founder - Full-Stack Developer & Product Designer | Live: cash-cram.vercel.app

- Gathered and evaluated user requirements from South African students to define needs-based budget presets (NSFAS, Bursary, Part-time, Gap Year)
- Designed a complete original visual identity and UI style guide - a Claymorphism design system (Nunito typography, coloured shadow depth, soft lavender-white palette)
- Built a spending intelligence engine, receipt scanner, budget breakdown widgets, streak calendar, and social leaderboard
- Built the backend: Firebase Authentication and Firestore data layer, Google Gemini API integration, and push notification infrastructure
- Tools: React 18, TypeScript, Vite, Tailwind CSS, Firebase Auth/Firestore, Google Gemini API, Vercel

Yandasm - Mental Wellness Platform for South African Students

Full-Stack Developer & Product Designer | Live: study-mate-dun.vercel.app/dashboard

- Designed and built a platform connecting students with trained peer counsellors via a booking and scheduling system and live one-on-one chat
- Designed a floating AI companion widget for always-on support between sessions, and a daily mood check-in feature with weekly visual history
- Designed a dedicated Emergency page surfacing South African crisis helplines (including 112 and the suicide prevention line) within one tap
- Tools: React, TypeScript, Tailwind CSS, real-time chat interface, scheduling/booking system

Fraud Detection System

Full-Stack Developer | Live: fraud-detection-blw97x5uesdprengweuci7.streamlit.app

- Developed an anomaly detection system for financial fraud using classification algorithms

- Designed and built an interactive ML dashboard for predictions and risk indicators
- Tools: Streamlit, Python, Scikit-learn, Pandas

Unemployment Risk Estimator (South African Youth)

Full-Stack Developer | Live: unemploymentriskestimatorforsouthafricanyouth-cr haczvjy cs8cpgr.streamlit.app

- Built an ML model predicting unemployment risk factors for South African youth
- Performed data preprocessing, feature engineering, and exploratory data analysis
- Designed and deployed an accessible, interactive Streamlit application
- Tools: Streamlit, Pandas, NumPy, Matplotlib, Scikit-learn

Football Score Predictor

Full-Stack Developer

- Built a predictive sports analytics model using historical match data
- Designed and deployed an interactive prediction interface
- Tools: Streamlit, Python, Pandas

ADDITIONAL PROJECTS

- Credit Card Approval Prediction Model
- Mental Health Stress During Loadshedding Analysis
- South African Economic Data Analysis
- Mindscope Optimizer

Full Portfolio: github.com/Unsolvedstar

CERTIFICATIONS

IBM Data Science Professional Certificate

- 6-course programme covering Python, SQL, data analysis, visualization, and data science methodology

Upskillist (2026)

- Cloud Architecture - Distinction
- Cybersecurity Certificate - Distinction

Codecademy (2025)

- SQL Data Analysis Path
- R Data Analysis Path
- Mathematics for Data Science
- Analyze Financial Data with Python Skill Path
- Software Engineering for Data Scientists Skill Path

Currently Learning

- Machine Learning (Python)
- Generative AI

EDUCATION

Eduvos Midrand Campus

BSc in Information Technology (Data Science)

Completed First Year (2023) — studies currently on pause

LEADERSHIP

Soccer Team Captain

- Led team to 4 championship victories
- Developed leadership, strategy, and teamwork skills

LANGUAGES

- English - Fluent
- Venda - Fluent
- Tswana - Fluent
- Sotho - Fluent
- Zulu - Conversational

REFERENCES

Available upon request